KAPASA MAKASA UNIVERSITY

ENTOMOLOGY

UNIT FOUR

TOPIC: DESCRIBING GROWTH AND DEVELOPMENT OF AN INSECT

Objectives: by the end of this unit, SSBAT;

- Explain embryonic growth
- Describe post-embryonic growth
- Examine types of metamorphosis.

EMBRYONIC GROWTH

Embryonic growth is the series of changes an embryo undergoes as it grows towards a mature organism. While embryogenesis is the process by which the embryo forms and develops. Both these definitions mean changes undergone from the egg to the final development of an insect (adult). They can formally be termed as an insect's life cycle.

A life cycle is the chain or sequence of biological events that occur during the life time of an individual insect. The cycle is usually considered to begin with deposition of the eggs and

end with the egg laying by the adult female insect. A life cycle lies in the boundaries of a single generation.

Related to the life cycle is the insect seasonal cycle of a species that occur over a period of one (1) year. By addressing the topic of seasonal cycles, we can more fully understand some of the ways in which insects can survive adverse times of the year.

An understanding of both life cycles and seasonal cycles for a pest species is required for proper management of crops and forests. The above knowledge will be useful in the following aspects:

a. Prediction of injury stages
b. Times when insects are vulnerable
c. Probable habits

Without this knowledge, development of effective and long term solutions to insect pest problems would be virtually impossible. Therefore, the evolution of insects from egg to adult vary from other animals such as reptiles, mammals and birds.

Understanding the life cycle of insects is equally important in deciding on the insect stage at which insects are susceptible to control method adopted, life of insects starts once eggs are laid by an adult.

POST – EMBRYONIC GROWTH

Moulting Process

This is the process of shedding the old cuticle and laying down of the new cuticle. It allows growth in larval stages i.e. each successive, instar gets bigger and allows metamorphosis in the transition from larva to adult. The exoskeleton of an insect is a non-living formation which does not change size and grow with the insect. As a result, it is necessary for the insect to shed its old exoskeleton to make way for a new larger one through a process called moulting. As the insect grows and the need for moulting arises, hormones are released in the insects' body which signal the process to begin. When juvenile insects reach a certain size, the exoskeleton becomes inviting for growth. A "stretch" signal is sent to the insects' brain, which results in the production of prothoracicotropic hormone (PTTH) by neurosecretory cells within the brain. PTTH is translocated and released from the corpus allatum glands into the haemolymph (insect equivalent of blood stream). Release of PTTH stimulates the prothoracic glands to release ecdysone into the haemolymph. A rise in levels of hydroxyecdysone (20E), triggers the start of the moulting cycle.

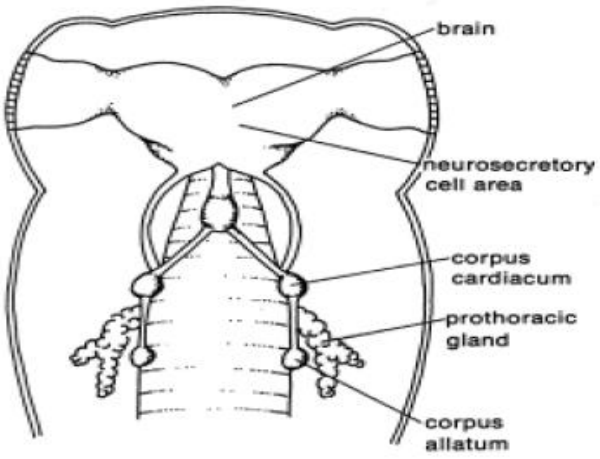


Figure 1. Major endocrine structures in insects.

Horn: 1978

Moulting fluid rich in the enzymes flows between the epidermis and old cuticle and dissolves the old cuticle. After the new cuticle (exoskeleton) is formed, the old cuticle cracks, the insect exists, the old cuticle and the new cuticle hardens and pigments.

The moulting process

1. Apolysis: The separation of the exoskeleton from the underlying skin cells. During this time, the insect will often become in active and may hide in order to protect itself during the coming moult.
2. The space produced by apolysis is filled by an inactive moulting fluid, which will later digest the bottom layer of the old exoskeleton.
3. A new cuticle layer is produced which will become the top layer of the new exoskeleton. This layer is often larger than the old one in order to allow for growth. Because it is soft when first formed, it is slightly wrinkled until the moulting process is complete and the body has filled in the new space.
4. The moulting fluid becomes active and digests the bottom layer of the old exoskeleton. This is why the shed moult appears thinner.
5. The protein components of the digested layer are re-absorbed and used in the process of building the new exoskeleton. The new cuticle layers are built until they are nearly fully formed.
6. Ecdysis: This is the actual process of shedding the exoskeleton. When the new exoskeleton is ready, the animal intakes air or water and contracts its muscles to swell its body until the old exoskeleton breaks along weak spots, often on the dorsal side. The arthropod then exits the old exoskeleton, leaving its moult behind.
7. The insect inflates itself into its new form, smoothing out the wrinkles in its soft new exoskeleton.
8. Sclerotization: The hardening of the new exoskeleton. Tanning agents are secreted through ducts onto the surface of the newly formed exoskeleton. These tanning agents create a chemical reaction, which cross-link the substances of the top layer of the exoskeleton into the hard shell we recognise.

METAMORPHOSIS

Metamorphosis refers to the way that insects develop, grow and change form. Metamorphosis actually infers “change”. The term is of Latin Greek origins:

Meta – change
Morphe – form/shape
Osis – the way it happens

Metamorphosis describes the series of changes through which an insect passes in its growth from egg stage (some insects, such as aphids, may produce eggs and/or give birth to live young) through the immature stages (Ex. Nymph, larva or pupa) to the adult stage.

1. Simple metamorphosis(Incomplete metamorphosis)
   Insects with simple metamorphosis have three life stages: egg, nymph, and adult. Adults and nymphs of these insects may live on the same foods. These insects may have wings, which if present, develop externally. Young ones are known as nymphs and there is no resting stage (pupa).
   An egg is the first stage of life. The egg hatches, and the animal is called a Nymph, which is the second stage of life. A nymph eats and grows. It eats the same kind of food the adults eat. In fact most nymphs look like miniature adults. As it grows, a nymph moults its outer covering and replaces it with a larger one. In most cases nymphs moult from four to eight times. The period of growth between each moult is called an instar. Each instar is much like a slightly larger version of the previous instar. After a final moult, a nymph becomes an adult, which is the third stage in its life. At that point, the insect has wings and is able to reproduce. Examples of insects in this category are:



2. Complete metamorphosis
   Insects with complete metamorphosis have four (4) life stages: egg, larva, pupa and adult. Larva are very different in form from the adult. The active immature stages are known as Larvae. More than 80% of insects, including flies, beetles, bees, butterflies and moths go through complete metamorphosis. The beginning of life for all insects is an egg laid by an adult insect. Inside the egg is a tiny animal called an embryo. The embryo develops inside the egg. The egg hatches and a larva emerges. A larva usually looks like a worm. The larva is the eating and growing stage. Some insects don't eat at all after this stage. Larva must moult, or shed their skin, a number of times as they grow. Caterpillars, grubs, maggots are all just larva stages of insects. After the larva has eaten and grown, the organism enters a resting stage when it does not feed or

move around much. This stage is known as the pupa stage. During which, many body changes take place as the insect is transforming. A complete metamorphosis occurs. New organs, muscles and body parts are developed. The pupa stage can last from a few days to many months. Eventually, an adult emerges from the pupa stage. Insects stop moulting when they reach their adult stage. The adult has all the parts that make it an insect. Three body sections, six legs, antennae and usually wings. Adult female insects continue the life cycle by reproducing and laying eggs.

